Evaluation of digital predistortion using the USRP N200 Software Defined Radio transceiver

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***Abstract*—This paper is oriented into domain of digital predistortion for power amplifier linearization. Its main contribution is a description of experimental evaluation of predistortion techniques using the commercial low-cost software defined transceiver USRP N200. The low-complexity predistorter based on the Look-Up-Table approach is used for verification. This type of predistorter has been chosen for the reason that although it cannot linearize the amplifier perfectly, its low complexity makes it suitable to be used in the transceivers based on the small FPGA's with limited free resources, e.g. in the low-end terminals.**

I. Introduction

In the current communication and wireless networking systems, the radio signals with non-constant envelope are mainly used in order to fulfill ever-increasing demands for high data rates. In order to gain better Power Amplifier (PA) efficiency, the amplifiers are often used close to the saturation point. Such a constellation leads to significant nonlinear distortion of the transmitted signals with undesirable effects on the increased bit error rate and spurious transmission into adjacent channels.

One of the solutions is a digital predistortion (DPD) of the power amplifier, usually performed in the complex baseband domain. The first real experiments with digital predistortion date back to 1980's [1]. The predistortion principle is based on the artificial distortion of the transmitted signal with the nonlinear function that models the inverse characteristics of the PA. The inverse function can be implemented in several ways differing in the complexity as well as in the performance of the linearization. The simplest way is based on the Look Up Table (LUT) or polynomial function approximation. These two approaches cannot compensate for the memory effects in the amplifier, significant especially for the wide-band input signals. In order to count for the memory effects, approaches like polynomials with memory, Volterra series or generalized memory polynomials have been proposed. The predistortion is often adaptive - the inverse function adapts to PA changes caused by changing carrier frequency, power or temperature variations.

Current research in the predistortion is often oriented into predistortion of mobile devices [2] or predistortion in MIMO systems, taking into account the crosstalk between various front-ends [3]. The search for optimal order of predistorters with memory is also currently investigated, see [4]. Note that besides the digital predistortion, the analog predistortion with adaptive mechanism can also be used [5]. The predistortion performance is affected by other RF impairments, such as IQ imbalances that can be corrected for, see [6] or filters. An interesting application of anti-aliasing filter pre-correction is shown in [7]. Digital predistortion is used not only for class AB or B amplifiers, but its application to switched mode PA's has also been examined [8] recently.

Digital predistorters are often implemented in the Field Programmable Gate Arrays (FPGA) [9] and first experiments using the Software Defined Radio platforms have been published, as in [10]. Currently, single-purpose chips for digital predistortion [11] or specialized IP cores for programmable devices [12] are also available. Unfortunately, these two solutions are not easy to use in the commercial software defined radios, moreover often equipped with small or medium-sized FPGA devices.

II. LUT-based digital baseband predistortion algorithm

The basic principle of adaptive digital predistortion for linearization of power amplifiers is shown in Figure 1. Here we consider the baseband predistortion, where a transmitted baseband signal is upconverted (in digital or analog form) to desired carrier frequency. A part of the transmitted signal is fed back to the baseband through a directional coupler (and attenuator) to have a reference signal for a predistorter adaptation.

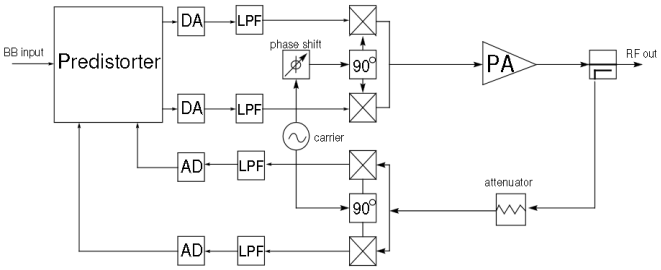


Fig. 1. Digital predistortion principle.

In the described experiment, we consider a complex gain LUT predistorter similarly as in [15]. The predistorter output z_p at time instant n is then

$$z_p(n) = z(n) f_{pre}(i), \tag{1}$$

where $f_{pre}(i)$ is a content of LUT entry i representing the inverse of the PA characteristics and $z(n)$ is a sample of

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modulated baseband signal. Note that in order to compensate for both amplitude and phase distortions, the LUT content is complex. The LUT can be addressed using either the instantaneous signal module or its instantaneous power. In order to get better linearization performance, the linear interpolation of the LUT entries is often done and we used it too. Recently, the advantage of the cubic interpolation has been described in [13]. The number of LUT entries can be significantly reduced in this case.

The adaptation of the predistorter has been done using an indirect-learning Recursive Least Squares (RLS) method published in [15]. In contrast to this sample-by-sample approach, we have slightly modified the algorithm to use it in the off-line block-by-block system. First the blocks of N data samples from the PA input and output are sampled, delay compensated (it's need is illustrated e.g. in [14]) and the predistorter LUT is computed from these blocks. Then the predistorter is applied to a new block of signal. In such a case we do not use any forgetting factor and the modulator output is used for the adaptation instead of the predistoter (that is not yet used in the adaptation phase) output. This leads to the adaptation formula:

$$f_{pre}(i) = \frac{\sum_{l=1}^{N} z(l) z_a(l)}{\sum_{l=1}^{N} |z_a(l)|^2}, \qquad (2)$$

where z_a is the PA output (divided by the desired gain of the system after predistortion) and z the digital modulator output (i.e. PA input). You can notice that such a method is very simple and thus suitable for implementation in programmable devices. Prior the implementation we analyzed the requirements on the bit representation of predistorter and its adaptation algorithm. First the minimal widths (b_{min}) were determined from the range of signals used for MATLAB model. Next, the simulation of the algorithm was run several times and additional bits were added in order to guarantee the adaptation stability (b_{sim}). Then the bit widths used for the implementation (the size of elementary FPGA Xilinx multiplier is 18x18 bits) were chosen. All the values are shown in Table I, with the signals corresponding to schematic in Fig. 2. Note that signals *numerator* and *denominator* correspond to the sums in the numerator and denominator in Eq. 2.

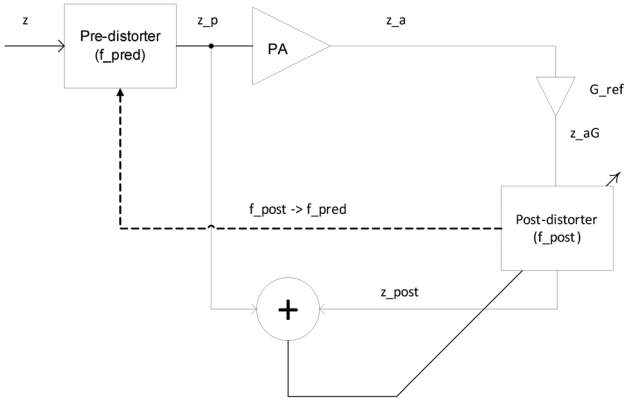


Fig. 2. Indirect learning architecture.

III. The USRP software defined radio

In order to test new techniques for communication systems, the use of an Universal Software Radio Peripheral (USRP) from Ettus company (now National Instruments) is very popular among researchers in both academia and industry. In many cases, the USRP is used in connection with a host PC, and the more advanced data processing is done in PC using either GNU radio, GNU radio companion graphical environment, Simulink or LabView as in [10].

TABLE I. LUT predistortion bit widths: minimal (b_{min}), simulated (b_{sim}) and used for implementation (b_{imp}).

signal	b_{min}	b_{sim}	b_{imp}
z	12	14	18
z_p	12	14	18
z_a	12	14	18
z_{aG}	12	14	18
G_{ref}	9	9	18
numerator	21	24	36
denominator	21	24	36
z_{post}	12	14	18

From among the variety of the USRP devices, we have used the USRP N200. Its main board is equipped with the Xilinx Spartan 3A-DSP 1800 device and dual 14 bits A/D converters with 100 MSa./s. sampling rate. The USRP can be used in connection with various front-end modules ranging from DC to 6 GHz frequency bands.

In the USRP, data from the antenna are received by the front-end module, converted to digital domain by the A/D converters and subsequently Digitally Down-Converted (DDC) to baseband. The USRP source code is ready to be modified by the custom Verilog code to be included in several positions of the transceiver chain. In the receiver, the user can access data from the front-end, input to the DDC, output of the DDC or baseband data. The transmitting chain is very similar. In our experiments we have accessed the final baseband data.

IV. Integration of the predistortion algorithm into FPGA

For the communication of the modules in the general USRP N2x0 design, an open source Wishbone bus is used. In this particular case the bus has 32-bit data width and 32-bit address space. There is one wishbone master (WB master) in the USRP design, which is the ZPU soft processor. The bus is configured to have 16 slaves, but some of the slave positions are not used. Memory map is quite complex because bus granularity is used. Our interface (implemented by TESLA a.s.) to the general USRP design is inserted between the DUC/DDC (Digital Up Converter/ Digital Down Converter) and the VITA RX/TX chain where the baseband IQ samples are processed (see Figure 3). By default, the USRP has two RX chains to be able to receive on two channels. As this functionality is not needed for the intended application, the second RX chain was removed from the design to save the FPGA resources.

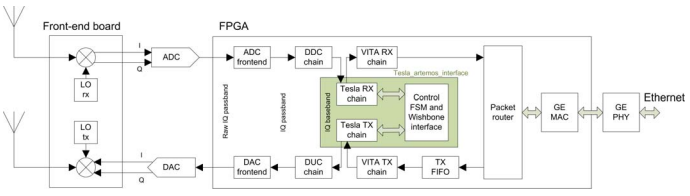


Fig. 3. Integration into USRP system.

TABLE II. FPGA RESOURCES FOR DPD AND ITS ADAPTATION.

	Slices	Slice F-F	DSP48	Block RAM
DPD	70	33	5	17
DPD+adaptation	1154	1682	29	22
USRP free/total	1548/16640	17796/33280	56/84	30/84

The LUT is addressed directly by the selected number of most significant bits of the DPD input squared module, see Figure 4. It avoids the complex search for the LUT entry corresponding to the instantaneous input signal at expense of the number of LUT entries being the power of 2. Each LUT entry contains 32-bit number with higher 16 bits for In-phase and lower 16 bits for Quadrature part. The LUT entry is then used according to Eq. 1.

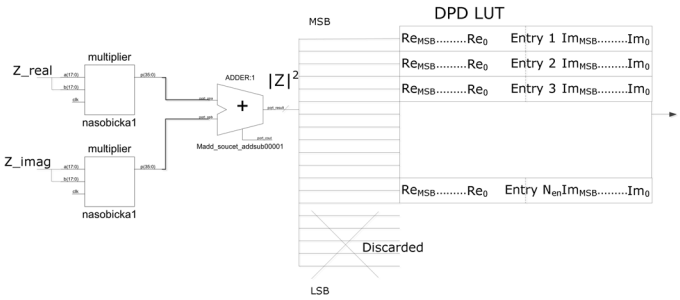


Fig. 4. Search for a LUT entry.

The adaptation of the predistorter coefficients has been done first off-line using developed MATLAB script according to Equation 2. For debug purposes, a simple design shown in Fig. 5 has been inserted into USRP FPGA image. It allows to fill the contents of a 4096-long block RAM with 32-bit wide entries (16 for I and 16 for Q part) from the host PC and run the transmission of the signal from this block RAM by the user command. Equivalently the received signal can be stored in the RX block RAM that can be subsequently read by the host PC and used for adaptation.

As a next step, the adaptation has been implemented in the USRP's FPGA device in cooperation with TESLA, a.s. company. The required resources for the digital predistorter (in the open loop) and the adaptation path together with the free resources in the USRP's FPGA and total Spartan 3A DSP resources are summarized in Table II. It is shown that the complexity of open-loop path is negligible. The complexity of the DPD with the adaptation is higher (mainly due to division in Eq. 2), but still affordable for this type of device.

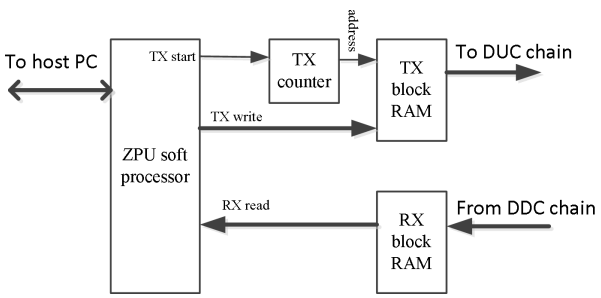


Fig. 5. Block RAM's in debug mode, used for signal capture.

V. MEASUREMENT SETUP AND RESULTS

A. Workplace setup

The workplace setup is shown in Fig. 6. The USRP N200 TX is used with the WBX front-end module, whose output is amplified with the PA working in nonlinear region. This PA to linearize was a monolithic InGap HBT amplifier GVA84+ from Mini Circuits with approximately 20 dB power at 1dB compression. At the PA output a directional coupler CP0603GN with -34dB coupling from AVX company has been used to get the attenuated part of PA output fed back to the USRP RX chain. The PA output is also monitored using a Rohde&Schwarz FSQ 3 spectrum and vector signal analyzer to get the output spectrum, IQ diagram and to analyze EVM. The USRP has been controlled via a control software in PC (a GNU radio companion plus custom programmed part). Using this software the sampling frequencies and RX/TX gains can be changed.



Fig. 6. Measurement workplace.

B. Measurement results

The PA has been fed up first with QPSK and 16-QAM signals (2 MHz bandwidth). The measured AM/AM characteristic is shown in black color in Figure 7. You can observe a dispersion of points in the characteristics caused possibly by the PA memory effects or incoherency of RX and TX oscillators. The effect of predistortion on the input signal is shown on the predistorter AM/AM characteristic (in cyan). The predistorter has been also tested with the LTE uplink signal with 3.84MHz bandwidth, generated by a generator SMU 200A with a LTE option. The spectra of the amplifier output with and without predistortion are shown in Figure 8 for 16QAM and LTE signals. You can observe around 10 dB reduction in the adjacent channels close to the main channel, i.e. in the regions where the filtration of the undesirable components is impossible. The application of predistorter also leads to EVM improvement (at working point from 3.3% to 1.7% RMS).

VI. CONCLUSION

The experiments with the digital baseband predistortion using the widespread commercial software defined radio system USRP N200 have been described in this paper. The bit widths necessary for predistortion implementation have been analyzed prior the experiments. It has been shown that even simple and low complexity Look Up Table-based digital predistorter can result in acceptable improvement in sense of adjacent

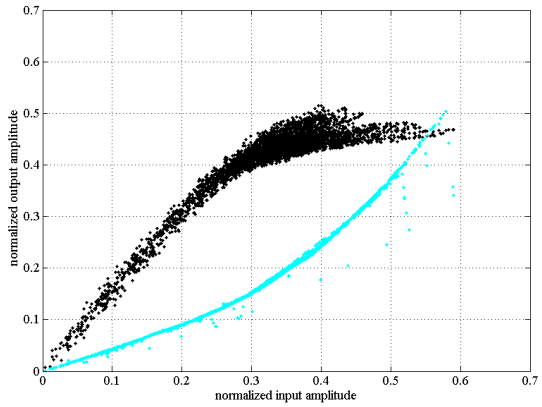


Fig. 7. Measured AM/AM characteristics of PA (black) and of DPD (cyan).

channel emissions and in-band distortions. According to the currently published results in [13], the performance can be further improved using the quadratic interpolation of LUT entries. Further work will be oriented to the evaluation of more advanced predistorters in the same workplace and in the optimization of adaptation implementation.

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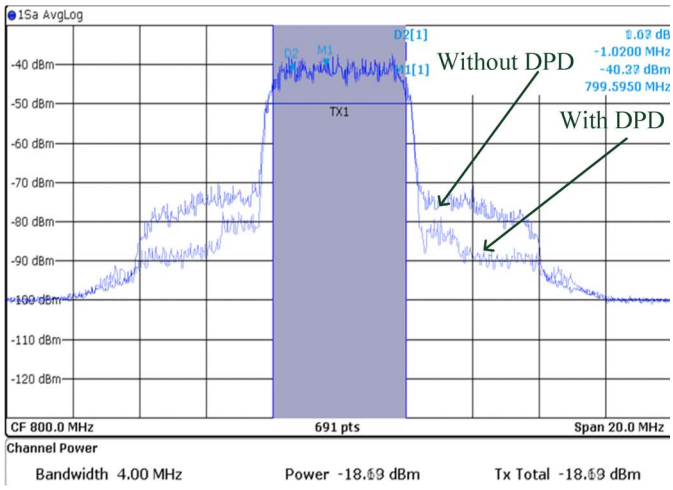


Fig. 8. Spectra of the PA output without and with predistortion for 16-QAM (top) and LTE (bottom) signals.

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